

History of the Enigma

The Enigma Machine was Germany's main cryptographic device during the Second World War. It was invented in 1919 by Dutchman, Hugo Koch. It was first produced commercially by Arthur Scherbius in 1923. The German government took an interest in the security it provided and acquired all rights to the machine and set about to adapt and change it to its specific, military needs. As the military adopted it, its use penetrated all levels of command from the front-line to ships, tanks, and planes (Bury).

The machine, unaltered, with the three rotor, reflector and 6 plug connector set-up was capable of 3,283,883,513,796,974,198,700,882,069,882,752,878,379,955,261,095,623,685,444,055,315,226,006,433,616,627,409,666,933,182,371,154,802,769,920,000,000,000 coding positions. Knowing that the deciphering of the Enigma without knowledge of the rotor configuration, and other factors, would require machine assistance that was had not been invented yet, the German's felt secure with the use of the Enigma, even if one was captured. Unfortunately for them, they did not always use proper communication security and this, along with the advent of the high-speed machinery needed to crack the codes helped to bring an end to the Third Reich.

The British worked to break the Enigma at Bletchley Park in England. The code breakers were organized into Huts. These Huts worked in pairs and were known only by their numbers for security reasons. Huts 3 and 6 worked together on deciphering messages from the German Army and Luftwaffe. Huts 4 and 8 worked on messages for the Navy. This history will focus mainly on the Navy's Enigma since that was the most difficult for the people at Bletchley to break.

The Navy's Enigma used many different ciphers. Each cipher had its own daily key that consisted of the rotor order, ring settings, plugboard connections and the ground setting (Erksine). Up until October, 5 1941, the primary U-Boat cipher was Heimisch or Dolphin. After that, Shark was the main cipher. The differences in the ciphers were aided by the differences in the hardware that they ran on. Dolphin ran on M3 and Shark initially ran on M3 and then switched to the new M4.

The Navy's original Enigma, M3, was of the same, three rotor design that was used by the army and luftwaffe but with the slight change of 3 additional rotors, VI, VII, and VIII. These 3 additional rotors were reserved for the Navy. The Navy also employed codebooks to shorten signals in order to protect against high frequency, land-based direction finding. The first of the two most important books were the Kurzsignalheft, the short signal book that was used for reports of sighting convoys, etc. The second book was the Wetterkurzschlüssel that was used for weather reports (Erksine). This gave Hut 8 a challenge.

The two main problems facing Hut 8 of Bletchley Park were:

- 1) The 8 rotors could be arranged in 336 different ways as opposed to the 60 for the army and luftwaffe. A bombe run using all the Navy's rotor combinations took five times longer than a run against the army or luftwaffe Enigma (Erksine).
- 2) 'Cribs', probable plaintexts that were used to program the Bombes, were non-existent until mid-1941 (Erksine).

The British were lucky enough to receive an Enigma and the first 5 rotors (I – V) from the Polish Cipher Bureau in 1939 before Germany invaded. A lot of credit goes to the Polish cryptanalyst Marian Rejewski who had reconstructed the wirings for the first 3 rotors in 1932 just by using mathematical techniques, and the wirings for the other 2 (IV and V) before the war began (Erksine). Rotors VI and VII were captured from the crew of U-33 on February 12, 1940 and the

last rotor was acquired in August of 1940. These acquisitions allowed the British to actually see the construction of the Enigma and to formulate a machine to combat it.

The famous mathematician Alan Turing invented the first Bombe, a machine designed to find the keys to ciphers in 1940. This allowed the British at Bletchley to break some messages starting in April 1940 and continuing through May and June. An improved, faster Bombe was invented and began service in August of the same year.

To aid the Bombes Hut 8 needed good cribs. Many cribs were provided by Hut 10, which broke manual weather ciphers. Even more cribs were created when Hut 8 received an 1940 copy of the Wetterkurzschlüssel. This allowed Hut 8 to break U-boat weather signals that provided the cribs.

But even with the new, faster Bombes and the cribs from Hut 10, Bletchley had to live with reading delayed Dolphin messages until June and July 1941 when keys captured from weather ships became available. From August of 1941 through until the end of the war, Hut 8 was able to read Dolphin signals with little delay.

The Germans however were not sitting idly by. On February 1st, 1942 the new M4 came online on Triton (named Shark by Bletchley), a special cipher used by Atlantic and Mediterranean U-boats (Erksine). The M4 used 4 rotors instead of 3 making Bletchley unable to read and Shark messages for over 10 months. However, the M4's 4th rotor was not interchangeable with any of the 8 original rotors. So the 4th rotor, Beta, increased the M4's power by 26 but did not change the number of different rotor configurations from the M3's 336. This was not M4's only problem. At one configuration of the 4th rotor, M4 emulated M3 and all U-boats used this configuration when encrypting the short weather reports. This, coupled with the seizure of the second edition of the Wetterkurzschlüssel from U-559 on October 30, 1942 which gave Hut 8 cribs, led to consistent, yet delayed decryption of weather broadcasts. These broadcasts allowed the British to locate and avoid the U-boats saving many ships and men between December 1942 and January 1943.

However, this success was short-lived. On March 10, 1943 a new version of the Wetterkurzschlüssel took effect which deprived Hut 8 of any cribs. But using short signal sighting reports as cribs, Hut 8 broke Shark again on March 19, 1943. The Kurzsignalheft short sighting reports used the M4 in M3 emulation mode as well and the Kurzsignalheft had also been seized from U-559 (Erksine). They were able to read Shark signals for 90 out of 112 days until June 20, 1943

To combat the new M4 the British and US Navy introduced their 4-rotor Bombes into service in June and August 1943. This initially allowed them to break Shark keys in 26 days. From September 1943 on however, they were able to break the Shark keys in less than 24 hours on average.

Some notes

Dolphin and most of the 14 other ciphers that were used throughout the war by the Navy consisted of a Allgemein (general) and Offizier keys. Offizier messages were first encrypted with plugboard settings from a monthly keylist and then were encrypted with the Allgemein key. Some of the ciphers also had a Stab (staff) keys which were, like the Offizier keys, encrypted twice. But the Stab keys had their own settings. Offizier messages usually took a week to decrypt at Bletchley (Erksine).

To read more on how the Enigma was solved, read Keith Jones' paper located at:
<http://www.cs.rit.edu/~kmj9907/enigma.html>